Experiment No 14.2

```
// TCP Server - Reverse String
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <winsock2.h>
#pragma comment(lib, "ws2_32.lib")

int main() {
   WSADATA wsa;
   char buff[512], str1[512], c;
   SOCKET SocketFD, connectFD;
   int len, i, k, a;
   struct sockaddr_in stSockAddr, cli_addr;

   if (WSAStartup(MAKEWORD(2, 2), &wsa) != 0) {
      perror("can not start winsock"); exit(EXIT_FAILURE);
   }
   SocketFD = socket(AF_INET, SOCK_STREAM, 0);
   if (SocketFD == INVALID_SOCKET) {
      perror("can not create socket"); exit(EXIT_FAILURE);
   }
   memset(&stSockAddr, 0, sizeof(stSockAddr));
   stSockAddr.sin_family = AF_INET;
   stSockAddr.sin_port = htons(1504);
   stSockAddr.sin_addr.s_addr = INADDR_ANY;

   if (bind(SocketFD, (struct sockaddr*)&stSockAddr, sizeof(stSockAddr)) == SOCKET_ERROR) {
      perror("can not bind"); closesocket(SocketFD); exit(EXIT_FAILURE);
   }
   listen(SocketFD, 2);
   printf("Server listening on port 1504...\n");

   a = sizeof(cli_addr);
   connectFD = accept(SocketFD, (struct sockaddr*)&cli_addr, &a);
   if (connectFD == INVALID_SOCKET) {
      perror("can not connect"); closesocket(SocketFD); exit(EXIT_FAILURE);
   }

   printf("\nReceived String: ");
   memset(buff, 0, sizeof(buff));
   recv(connectFD, buff, 512, 0);
   printf("%s", buff);

   len = strlen(buff);
   for (i = len - 1, k = 0; i >= 0; i--, k++) {
      c = buff[i];
      str1[k] = c;
   }
   str1[k] = '\0';

   printf("\nReversed String: %s\n", str1);
   send(connectFD, str1, 512, 0);

   closesocket(connectFD);
   closesocket(SocketFD);
   WSACleanup();
   return EXIT_SUCCESS;
}

// TCP Client - Reverse String
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <winsock2.h>
#pragma comment(lib, "ws2_32.lib")
```

```
int main() {
   WSADATA wsa;
   char buff[512];
   SOCKET SocketFD;
   struct sockaddr_in stSockAddr;

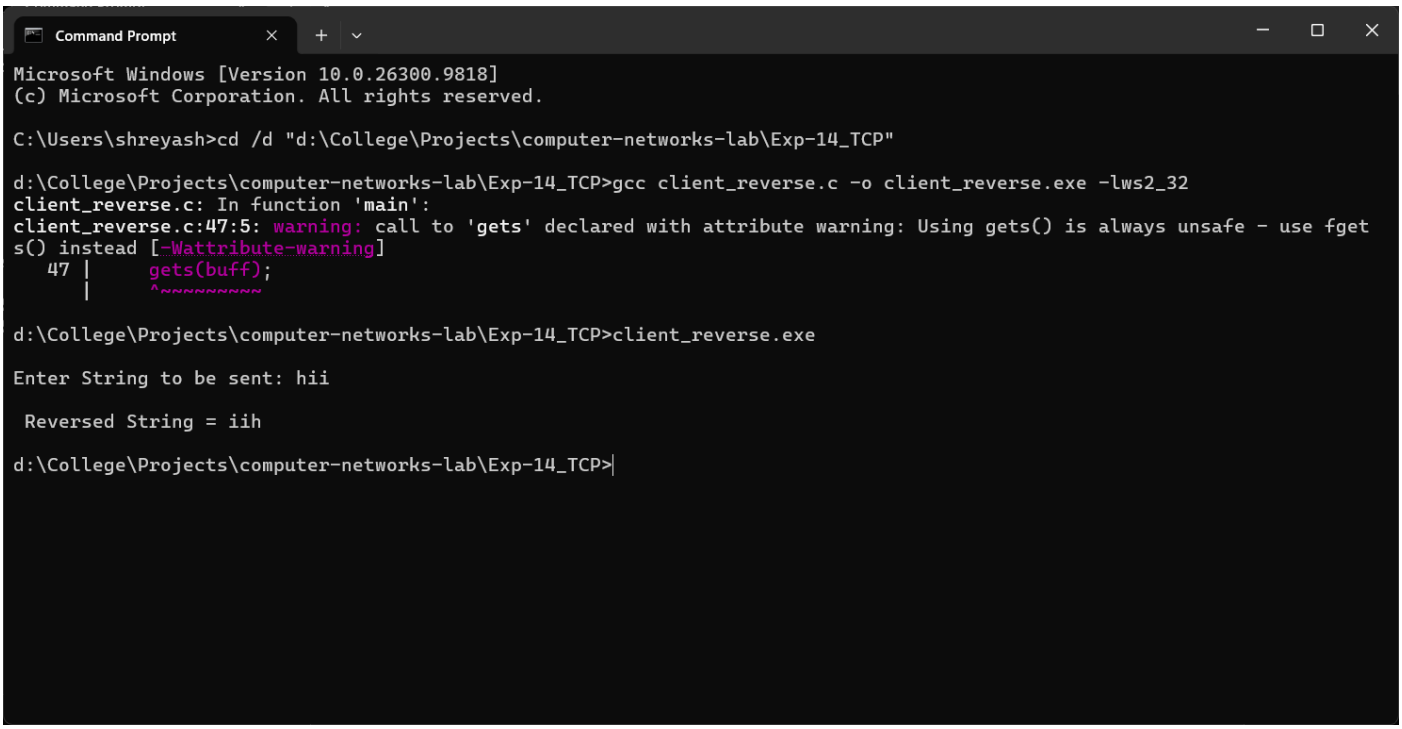
   if (WSAStartup(MAKEWORD(2, 2), &wsa) != 0) {
      perror("can not start winsock"); exit(EXIT_FAILURE);
   }
   SocketFD = socket(AF_INET, SOCK_STREAM, 0);
   if (SocketFD == INVALID_SOCKET) {
      perror("can not create socket"); exit(EXIT_FAILURE);
   }
   memset(&stSockAddr, 0, sizeof(stSockAddr));
   stSockAddr.sin_family = AF_INET;
   stSockAddr.sin_port = htons(1504);
   stSockAddr.sin_addr.s_addr = inet_addr("127.0.0.1");

   if (connect(SocketFD, (struct sockaddr*)&stSockAddr, sizeof(stSockAddr)) == SOCKET_ERROR) {
      perror("can not connect"); closesocket(SocketFD); exit(EXIT_FAILURE);
   }
   printf("\nEnter String to be sent: ");
   gets(buff);
   send(SocketFD, buff, 512, 0);

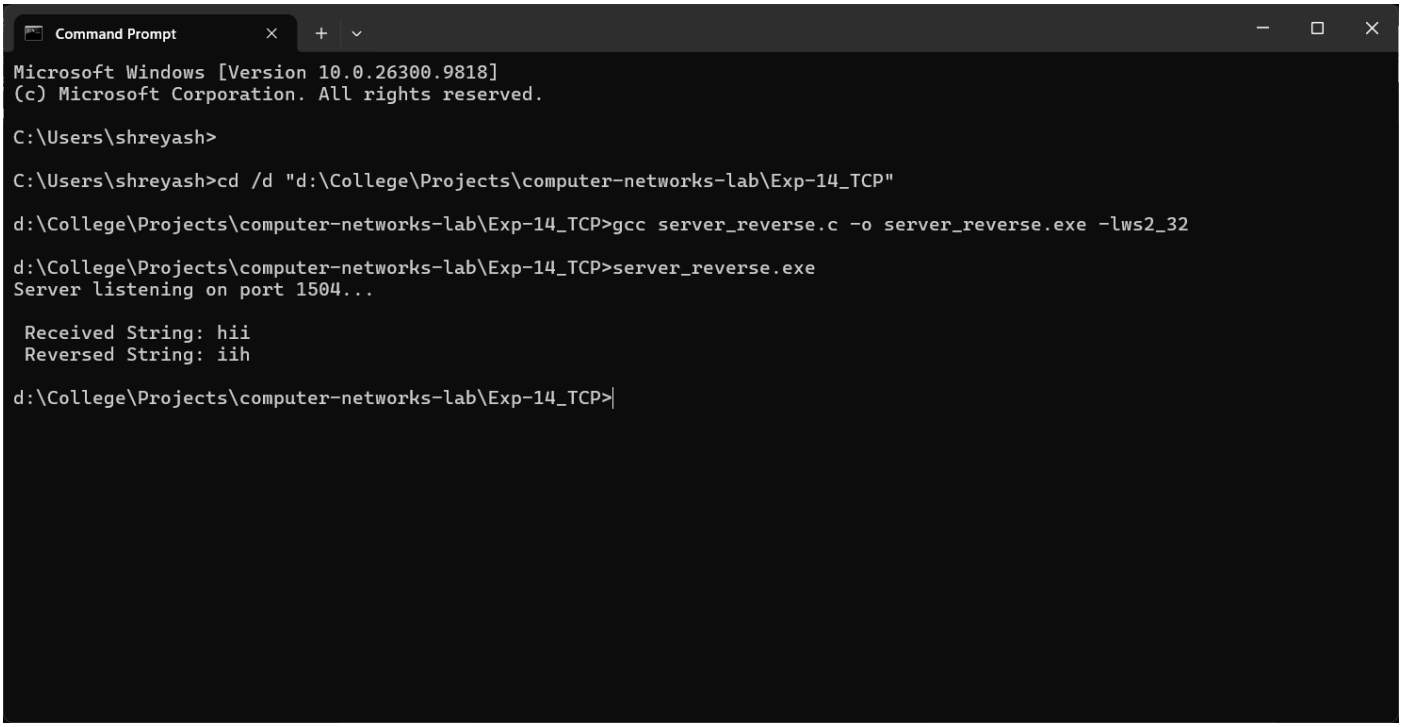
   memset(buff, 0, sizeof(buff));
   recv(SocketFD, buff, 512, 0);
   printf("\nReversed String = %s\n", buff);

   closesocket(SocketFD);
   WSACleanup();
   return EXIT_SUCCESS;
}
```

Output :
Client :



Server :


Command Prompt
Microsoft Windows [Version 10.0.26300.9818]
(c) Microsoft Corporation. All rights reserved.
C:\Users\shreyash>
C:\Users\shreyash>cd /d "d:\College\Projects\computer-networks-lab\Exp-14_TCP"
d:\College\Projects\computer-networks-lab\Exp-14_TCP>gcc server_reverse.c -o server_reverse.exe -lws2_32
d:\College\Projects\computer-networks-lab\Exp-14_TCP>server_reverse.exe
Server listening on port 1504...
Received String: hii
Reversed String: iih
d:\College\Projects\computer-networks-lab\Exp-14_TCP>